

Lecture 11 – AI Ethics and Future Development

INT104 Artificial Intelligence

AI Ethics and Future Development

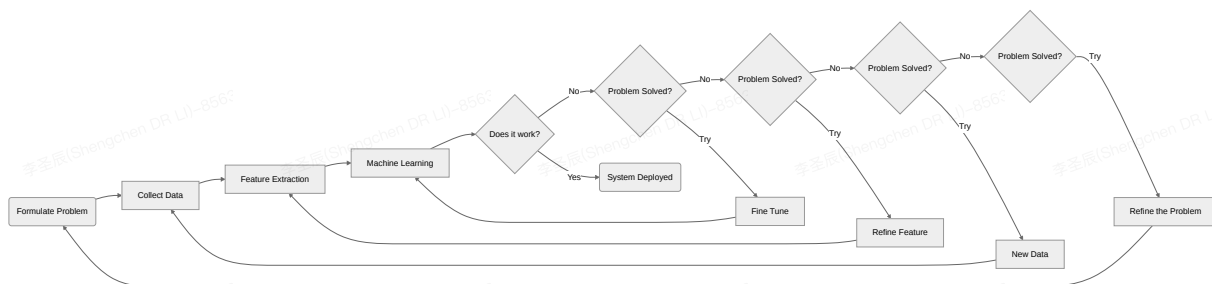
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AI Systems



Concerns

Problems

- Data Privacy
- Data Collection (Bias)
 - Data Availability
- Algorithm Bias
- Evaluation Bias
- System Security
- False Media
- Power Consumption of the System

Practice

- Anonymisation (Really Work?)
- Know Your Population (Particularly Background Information)
- Know Your Algorithm
- Know Your Preference
- Improve Robustness of the System
- Deepfake Detection
- Algorithm Simplification

Learning with Anonymisation (Federated Learning)

- Ideally, the data could be managed in a centralised manner
- However, the transmission of data take the risk of data leaking
- The data is stored in local machines hence could suffer from
 - inconsistent feature definition (or data collection)
 - inconsistent data quality
 - inconsistent feature distribution

Data Collection Bias



Think about this: what is the main purpose of machine learning?

- Essentially, machine learning attempts to set up a mapping between two feature spaces (e.g. label space and data space)
- The label space and the data space needs to be described as a probability distribution



Next, think about how the training dataset works?

- Hopefully, the training dataset could estimate the probability distribution of the real probability distribution.
- But the training dataset never has the same probability distribution with the ground truth (data bias)

Data Collection Bias

- How can we form a good dataset
 - Cover enough variety
 - Have enough number
- What if we cannot cover all kinds of varieties?
 - Different circumstances means different domains
 - Machine learning with domain shifts (a kind of transfer learning)
 - ◆ Domain generalisation
 - ◆ Domain adaptation

- What if we do not have enough data?
 - Few-shot learning / First-shot learning / Zero-shot learning

Algorithm Bias

- When you have used a machine learning algorithm, you are introducing assumptions:
 - GMM: all clusters can be approximated by Gaussian distribution
 - SVM: there is a feature space that may classify data with a linear boundary
 - Naive Bayes: the impact of factors is independent with each other
- Thus you need to make the introduced assumptions work well with your problems

Evaluation Bias

- Recall the knowledge related to confusion matrix
 - Accuracy: related to overall performance
 - Precision: quality of positive prediction
 - Recall: ability of identify positive class
- Recall what distance measurement we have?
 - Euclidean distance (measure absolute distance)
 - Cosine distance (measure directional distance)
 - Correlation distance (measure temporal distance)
- Different evaluation methods pay attention to different impacts

System Security

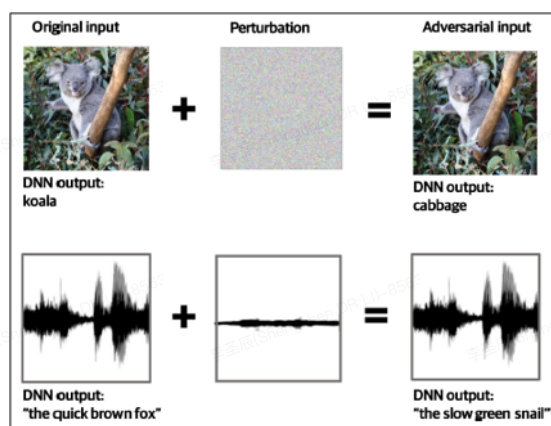


Figure 1-3. Adversarial perturbation applied across an image to fool an image classifier and across audio to fool a speech-to-text system

- AI systems can be attached: the following actions could mislead an AI system:
 - Perturbation: add minor invisible data noise
 - Data poisoning: add minor data into a dataset
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Deepfake

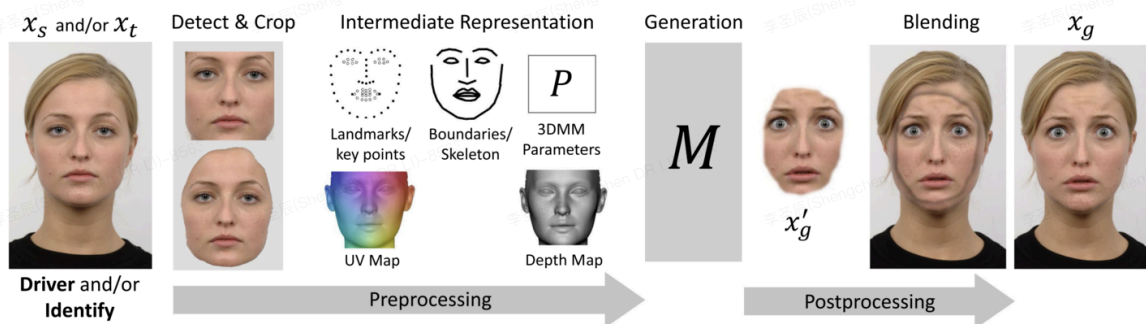


Fig. 5. The processing pipeline for making reenactment and face swap deepfakes. Usually only a subset of these steps are performed.

Power Consumption (Environment Friendly)

i AlphaGO has beaten Jie Ke. Is that true?

The power that Jie Ke has consumed $\rightarrow$ 500 KCal $\approx$ 0.0005 kWh

The power that AlphaGO has consumed $\rightarrow$ 60 kWh

As a result, it is necessary to simplify machine learning algorithms wherever possible (especially for deep learning system).

Future Challenges

Problems

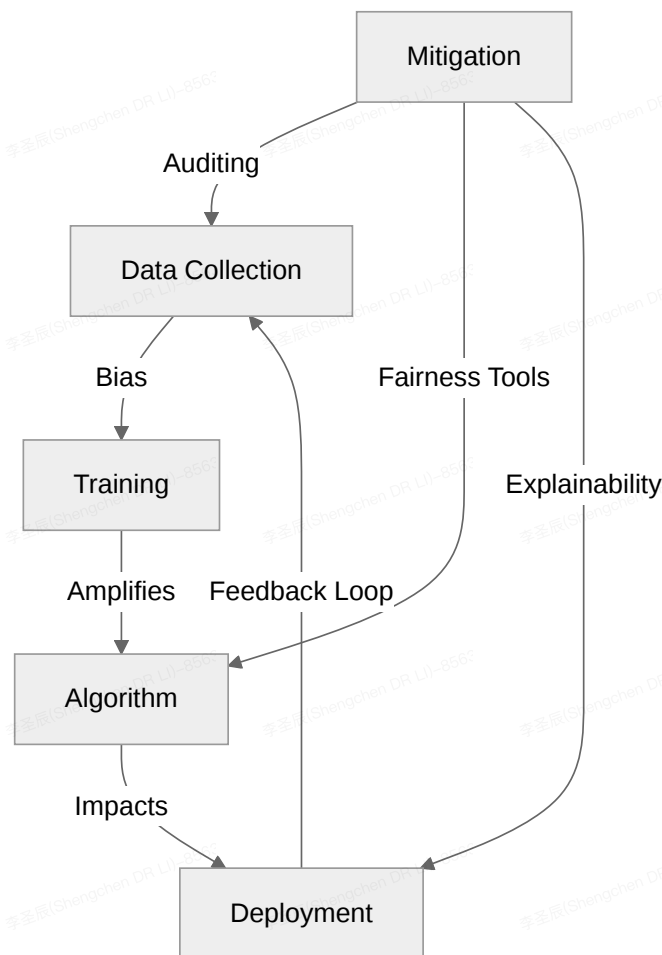
- Feature distributions are not same given different context
- Data Imbalance
- Low Data Availability
- New Classes in Validation Domain
- Auto Adaption in New Environment

Techniques

- Domain generalisation / Domain adaptation
- Transfer learning
- Few-shot learning / Outlier detection
- Few-shot learning / First-shot learning
- Openset / Zero-shot learning

Future Directions

- **Explainable AI (XAI):**
 - Interpretability to audit bias sources (e.g., LIME, SHAP).
- **Regulatory Frameworks:**
 - EU AI Act, Algorithmic Accountability Bills.
- **Ethics-by-Design:**
 - Embed bias checks at every AI lifecycle stage (problem formulation to deployment).



The question is not whether machines can think, but whether humans can design machines that think ethically.